Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/12/2005
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Project Title:	Induction Of Bystander Effects By High LET Radiation In Cells And In Tissue Models		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Element:	(1) Cell & Molecular Biology		
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Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	02114-2696	**Congressional District:**	9
Comments:			
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
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Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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With charged particles such as alpha-particles and low energy protons, significant differences from direct DNA damage-mediated responses have been observed in a number of biological systems. Occurrence of these non-targeted effects, including bystander effects and genomic instability, calls into question the use of a simple linear extrapolation of cancer risks from high dose data to low doses. Two important considerations are of particular relevance to NASA. First, the cancer risks from high doses of high LET radiations such as heavy ions and of energetic protons are not well understood, making it difficult to extrapolate to low doses. Second, the occurrence of bystander effects in cells exposed to heavy ions or energetic protons, as found in space, has not been evaluated to date. The overall goals of this project are to test the hypothesis that heavy ions and energetic protons, as encountered in space, efficiently cause bystander effects in neighboring cells not directly hit by the radiation and to study the mechanisms involved. Hence, the studies specifically address the Research Emphasis for FY2003 of “Molecular radiation biology of carcinogenesis”, as described

Task Description:	in the NASA Research Announcement. The studies will aid in assessing the risks to astronauts of carcinogenesis as a result of exposures to radiation in space. There are four aims. 1. Determine induction of bystander effect(s) by heavy ions compared to X-rays, protons (low and high energy) and alpha particles. Cultured human diploid fibroblasts and keratinocytes will be used with endpoints specific for each cell type and including production of micronuclei, apoptosis, clonogenicity and gene expression. Dose response curves for the induction of bystander effects will be obtained for a range of particles and energies available at the Brookhaven National Laboratory (BNL) NASA Space Radiation Laboratory (NSRL). Results with heavy particles and high energy protons will be compared with those obtained using full field irradiations with X-rays, protons and alpha particles. We will test the hypothesis that heavy ions and high energy protons effectively induce bystander effects in neighboring cells not traversed by the particles. 2. Test the hypothesis that heavy ion-induced bystander effect(s) can be modified by exposure of cells to reactive oxygen species (ROS), an overlying oxidative stress. Cells will be exposed to acute and chronic ROS, before, during and/or after exposure to the heavy ions, and bystander effect(s) will be evaluated. 3. Test the hypothesis that exposure to low LET radiations alters the heavy ion-induced bystander effect(s). Cells will be exposed to very low doses of low LET radiation, before and/or after exposure to the heavy ions, and bystander effect(s) will be evaluated. 4. Begin development of 3-D tissue models for assessment of bystander effects from heavy ions and energetic protons. A long-term goal of this work is to investigate the role of bystander effects in tissues, composed of mixed cell types in 3-D arrangements. These studies will test the hypothesis that bystander effects are also seen in tissues. The overall goals of this project are to test the hypothesis that heavy ions and energetic protons, as encountered in space, efficiently cause bystander effects in neighboring cells not directly hit by the radiation and to study the mechanisms involved. The studies specifically address the Research Emphasis for FY2003 of “Molecular radiation biology of carcinogenesis”, as described in the NASA Research Announcement. The studies will aid in assessing the risks to astronauts of carcinogenesis as a result of exposures to radiation in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research should provide added insight into mechanisms involved in radiation-induced carcinogenesis and cell killing. Such new insight may be useful for developing new approaches for prevention or treatment of cancer.
Task Progress:	At the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory we have conducted bystander response experiments all four times beam time has been available since this project was funded in June 2003. To date, our studies have concentrated on addressing aim 1 using 1 GeV/amu Fe (4 runs), 1 GeV/amu Ti (2 runs), 290 MeV/amu C (3 runs) and 1 GeV/amu protons (2 runs) in AG01522 normal human fibroblasts and immortalized human keratinocytes. The following preliminary observations have been made with the human fibroblasts: ·Using micronuclei (MN) formation, p21 expression and clonogenic survival as endpoints, we have observed a significant increase in response with dose in the directly irradiated cells after exposure to iron ions, with RBE values of 1.5 for MN and p21 and an RBE of 2.5 for survival compared to X-rays. Of note, is that the expression of p21 was delayed in cells irradiated with heavy ions compared to X-rays or protons (full expression by 5 hours with X-rays, but not until 40 hours or later with heavy ions) when expression was assayed by in situ immunofluorescence, but when expression was assayed by western blotting, the delay was not seen. At this time we do not understand the discrepancy, and additional studies are underway. Preliminary cell cycle analyses show Fe ions cause a longer G2/M delay (up to 50 hours) than X-rays. ·The RBEs are lower for Titanium, Carbon and Protons than for Iron. For example, for colony formation, the RBEs are 2.0, 1.0 and 1.2, respectively for Ti, C and protons. To date, our data suggest that different endpoints may have different correlations with LET, but additional repeats of experiments are still needed. ·In AG0 cells, a bystander effect was seen (2-3-fold increase compared to unirradiated controls) for MN after Fe irradiation, an increase similar in magnitude to that seen with X-rays. The increase in damage in bystander cells is independent of the dose to the hit cells between 0.1 and 2 Gy. The clonogenic assay shows that the cloning efficiency of bystander cells was reduced about 20% with Fe radiation, which is similar to the results obtained with X-rays. These results with Fe have been seen in all four experiments to date and are nicely reproducible. ·In two experiments, there does not appear to be a bystander micronuclei response after Ti or proton irradiation. This is surprising, as we expected Ti to cause similar responses as Fe, and protons to be similar to X-rays, and both Fe and X-rays cause bystander MN induction. Clearly, additional studies with Ti and protons are needed. ·Also surprisingly, there appears to be little bystander effect with iron, titanium or carbon for p21 expression, in contrast to our results with X-rays. In the human keratinocytes, the following preliminary results have been obtained: ·Keratinocytes irradiated with Fe or Ti show substantial direct effects for micronuclei and p21 induction (e.g., 30% MN after 3 Gy of Fe). Additionally, we have noted a significant level of formation of chromatin bridges in these cells after heavy ions and X-rays, an effect that is very small in fibroblasts. This is a unique cell-specific finding that requires follow-up. ·Although the keratinocytes show a significant bystander effect after X-rays (e.g., 2-3-fold increase in MN), much to our surprise there appears to be no bystander effect for MN and only a small effect for a p21 endpoint after heavy ions. The reason for this difference is unclear, but it may be related to the induction of cell cycle arrest by heavy ions (see next bullet). ·After heavy ion irradiation, the “hit” keratinocytes show a substantial delay in p21 expression (as do the fibroblasts). We have investigated this further in one experiment by doing cell cycle analysis, with results suggesting a large and prolonged G2 arrest in these keratinocytes. This finding needs confirmation. ·For reasons that are unclear to us, our data with protons (1 MeV/amu) in keratinocytes show only a small effect of direct irradiation (far fewer MN than after X-rays) and no bystander effect. However, these data have been less reproducible than with other ions, and clearly need repeated experiments. Studies have also begun on aim 2 using the human fibroblasts. These studies show that part of the X-ray-induced bystander effect (endpoints of micronuclei and p21 expression) is removed by addition of superoxide dismutase or catalase. This is consistent with involvement of ROS in transmittal of the bystander effect. Lastly, preliminary studies have been initiated on aim 4, by assessing the bystander effect between AG01522 human fibroblasts and the human keratinocytes after X-rays. Bystander signaling was seen from the AG01522s to the keratinocytes, with a two fold increase in both p21 and MN in bystander keratinocytes after doses as low as 0.1 Gy to

	the fibroblasts. Further investigation is currently ongoing on keratinocytes inducing bystander effects in the AG01522 cells. In summary, we have made a number of interesting, novel findings in both directly irradiated and bystander cells. These initial studies need extensive follow-up experiments.
Bibliography Type:	Description: (Last Updated: 11/13/2018)
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Abstracts for Journals and Proceedings	Held, K. D.; Yang, H; Asaad, N. "Comparison of the radiation-induced bystander effect in fibroblasts after treatment with X-rays or heavy ions" 3rd International Workshop on Space Radiation Research and 15th Space Radiation Health Investigators' Workshop, Port Jefferson, NY 2004 , May-2004
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